

GRIFFITHS POSITIVITY DOES NOT IMPLY POSITIVITY OF THE TOP CHERN FORM

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ABSTRACT. For every integer $r \geq 9$, we construct a smooth Griffiths-positive Hermitian metric on $\mathcal{O}_{\mathbb{P}^r}(1)^{\oplus r}$ whose top Chern form is pointwise negative on a nonempty open set. This gives counterexamples on compact projective manifolds to Griffiths' conjecture on the positivity of Chern–Weil forms.

1. INTRODUCTION

1.1. Chern–Weil positivity and the main theorem. Let X be a complex manifold of dimension n , and let (E, h) be a Hermitian holomorphic vector bundle of rank r on X . The metric h is Griffiths positive if the curvature R_h of its Chern connection satisfies

$$\langle R_h(u, \bar{u})v, v \rangle_h > 0$$

for every $x \in X$ and all nonzero $u \in T_x^{1,0}X$ and $v \in E_x$. We normalize the Chern forms by

$$\det \left(I_r + s \frac{i}{2\pi} R_h \right) = \sum_{q=0}^r s^q c_q(E, h).$$

Assign weight q to c_q . Fulton and Lazarsfeld [12] identified the cone of homogeneous positive polynomials considered by Griffiths with the nonzero polynomials

$$P = \sum_{\substack{\lambda \vdash k \\ \ell(\lambda) \leq r}} a_\lambda s_\lambda, \quad a_\lambda \geq 0,$$

where s_λ is the Schur polynomial in the r formal Chern roots, $\ell(\lambda)$ is the number of parts of λ , and the coefficients are not all zero. We use the convention in which c_j is the j th elementary symmetric polynomial in these roots; thus $s_{(1^r)} = c_r$.

In 1969, Griffiths [14, p. 247] proposed what is now called Griffiths' conjecture on the positivity of Chern–Weil forms. It asserts that, if X is a compact complex manifold of dimension n and (E, h) is a Griffiths-positive Hermitian holomorphic vector bundle of rank r on X , then for every integer k with $1 \leq k \leq n$ and every nonzero positive polynomial $P \in \mathbb{Q}[c_1, \dots, c_r]$ of weighted degree k , the Chern–Weil form

$$P(c_1(E, h), \dots, c_r(E, h))$$

is Hermitian positive at every point of X . Here a (k, k) -form is Hermitian positive if at each point it can be written as $\sum_j i^{k^2} \xi_j \wedge \bar{\xi}_j$ for $(k, 0)$ -covectors ξ_j . Equivalently, the positive cone spanned by the Schur forms of (E, h) consists of Hermitian positive differential forms, with Hermitian positivity understood as in Fagioli [10, Remark 1.4 and Proposition 1.7].

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The theorem below gives a counterexample with $X = \mathbb{P}^r$, $k = n = r$, and $P = c_r$. Here c_r is a Schur polynomial, and all standard positivity cones coincide in top degree; the pointwise negativity in the theorem therefore contradicts the conjectured Hermitian positivity.

Theorem 1.1. *For every integer $r \geq 9$, the holomorphic vector bundle $E_r := \mathcal{O}_{\mathbb{P}^r}(1)^{\oplus r}$ admits a smooth Hermitian metric h_r which is Griffiths positive on $\mathbb{P}^r$ and whose top Chern form $c_r(E_r, h_r)$ is pointwise negative on a nonempty open subset of $\mathbb{P}^r$.*

Thus Griffiths' conjecture on the positivity of Chern–Weil forms fails on compact projective manifolds, even for a direct sum of an ample line bundle. The smallest rank for which the top Chern form of a Griffiths-positive metric can fail to be weakly nonnegative remains open.

Griffiths [14] proved positivity of the second Chern form in rank two. Fagioli [10] established weak nonnegativity of the second Chern form in arbitrary rank under Griffiths semipositivity and of $c_1 c_2 - c_3$ in rank three. Wan [21] proved weak positivity of the third Chern form in arbitrary rank and positivity of all Schur forms for rank-three Griffiths-positive bundles over threefolds. Guler [15] proved positivity of the signed Segre forms for Griffiths-positive bundles on projective manifolds. Together, these results give affirmative cases in each of the first three Chern degrees; the counterexample below arises from a rank-nine construction.

Stronger curvature hypotheses also ensure positivity of characteristic forms. The curvature condition now called Bott–Chern nonnegativity gives nonnegative Chern forms by Bott and Chern [4]; Li [18] extended the conclusion to Schur forms for Bott–Chern nonnegative metrics on compact complex manifolds. Finski [11] proved positivity of all Schur forms under Nakano or dual Nakano positivity. Wan [20] obtained weak positivity under strongly decomposable positivity of type I and positivity under strongly decomposable positivity of type II. Thus the failure in Theorem 1.1 already occurs under Griffiths positivity.

The numerical theory concerns Chern numbers and cohomology classes on projective varieties. Bloch and Gieseker [3] established numerical positivity of the Chern classes of ample vector bundles on projective varieties. Fulton and Lazarsfeld [12] proved that the positive polynomials described above are precisely the homogeneous numerically positive polynomials for ample vector bundles. The theorem below concerns the sign of a Chern–Weil representative at each point, for a specified metric.

1.2. Reformulation and construction strategy. The pointwise curvature problem has a matrix formulation in terms of positive linear maps. For $n \geq 1$, write $M_n = M_n(\mathbb{C})$. Transpose and conjugate transpose are denoted by T and $*$, respectively. A complex-linear map $\Phi : M_n \rightarrow M_q$ is positive (respectively, strictly positive) if $\Phi(H) \geq 0$ for every $H \geq 0$ (respectively, $\Phi(H) > 0$ for every nonzero $H \geq 0$). By spectral decomposition, positivity is equivalent to the rank-one test

$$v^* \Phi(uu^*)v \geq 0 \quad (u \in \mathbb{C}^n, v \in \mathbb{C}^q). \quad (1.1)$$

Finski [11, Theorem 1.3] reduced the universal problem of weak positivity of Schur forms to a finite-dimensional problem for positive linear maps. When the source and target matrix spaces are both M_n and one considers the top Chern form in complex dimension n , the scalar takes the following form. For a matrix variable $Q = (q_{ij}) \in M_{r,s}(\mathbb{C})$, whose entries are independent complex variables, write

$$\partial_Q := \left(\frac{\partial}{\partial q_{ij}} \right)_{1 \leq i \leq r, 1 \leq j \leq s}.$$

For a square matrix variable, $\det(\partial_Q)$ is the determinant of this matrix of commuting differential operators. With $Y = (y_{ij})$, the scalar associated with a complex-linear endomorphism Φ

of M_n is

$$F_n(\Phi) := \det(\partial_Y) \det \Phi(Y). \quad (1.2)$$

For a Hermiticity-preserving map $\Phi : M_n \rightarrow M_n$, consider near the origin the Hermitian metric

$$h_\Phi(z) := I_n - 2\pi\Phi(zz^*).$$

At the origin, $\langle R^{h_\Phi}(u, \bar{u})v, v \rangle_{h_\Phi}|_0 = 2\pi v^* \Phi(uu^*)v$ for $u, v \in \mathbb{C}^n$, and its top Chern form is

$$c_n(\mathbb{C}^n \times \mathbb{C}^n, h_\Phi)|_0 = F_n(\Phi) dV_n|_0. \quad (1.3)$$

Here $dV_n = \bigwedge_{j=1}^n (i dz_j \wedge d\bar{z}_j)$. Thus the geometric problem leads to the construction of a strictly positive map for which F_n is negative.

The same formulation contains the known two- and three-dimensional cases. Finski [11, Proposition 1.4(c)] proved the required nonnegativity when $n = 2$. Wan [21, Theorem 0.3] proved it when $n = 3$. His argument first uses operator scaling to pass, without changing the sign of F_3 , to maps that are unital and trace preserving [21, pp. 8–9]. It then expresses this scalar through spherical moment identities involving traces of products of matrices and applies a symmetric spectral inequality [21, Lemmas 2.6 and 2.9 and Theorem 2.11]. In dimension four, the analogous expansion contains an additional sum of traces indexed by the 4-cycles, which complicates the extension of this spectral argument [21, Remark 2.12].

After constructing a rank-two ample bundle with no smooth Griffiths-semipositive metric [8], Du and Xie developed a measure-theoretic criterion for Griffiths-positive metrics on smooth projective varieties in [9]. Condition (iv) characterizes the existence of a Griffiths-positive metric by the absence of a nonzero positive matrix-valued measure annihilating the relevant Levi forms. Its finite-dimensional version produces, when separation fails, an atomic probability measure under which matrices formed from first jets have zero average [9, Proposition 2.1]. This formulation led the author to study the present positivity problem through expectations of matrix-valued data. Hoeffding's distribution comparisons for sums of independent Bernoulli variables [17], Darroch's bound relating their mean and mode [7], and Gnedin's extension of the mean-mode relation [13, Theorem 2] provided initial models for extracting sign information from expectations. The formula for $F_n(\Phi)$ called for a matrix distribution under which the two determinant expansions, with their row and column permutations and signs, can be evaluated together.

Complex Gaussian matrices supply such a law. If $G_n = (g_{ij})$ is an $n \times n$ matrix of independent standard complex Gaussian entries, normalized by $\mathbb{E}|g_{ij}|^2 = 1$, then

$$F_n(\Phi) = \mathbb{E}[\overline{\det G_n} \det \Phi(G_n)]. \quad (1.4)$$

This identity is a specialization of Bargmann's classical Gaussian integral representation of the apolar inner product [2]; Aldaz, Bravo, and Render [1, Theorem 7] give a modern polynomial formulation. Independence and the orthogonality of complex Gaussian monomials pair the coefficients of the two determinants in (1.4). Gaussian integration by parts then transfers conjugate factors to derivatives, allowing the terms selected by the construction to be expressed as weighted moments. Lemma 2.1 establishes these identities; the integration-by-parts formula is the complex polynomial counterpart of Stein's Gaussian identity [19, Lemma 1].

Wan's rank-three argument provides a closely related use of moment identities [21, Lemmas 2.6 and 2.9]. In rank three, the trace terms produced by the cycle expansion combine into elementary symmetric polynomials of the positive matrix associated with each sphere vector, so a spectral inequality proves nonnegativity. Our use of the Gaussian pairing in (1.4) retains the simultaneous row and column correlations in the two determinant expansions. Orthogonality and integration by parts express their contribution through weighted moments, which the

Cayley identities evaluate; these moments are not controlled by the rank-one positivity tests, and for a suitable map the rank-nine top Chern coefficient is negative.

Equation (1.4) characterizes the difference between the two positivity properties at the level of one curvature tensor. Griffiths semipositivity is nonnegativity of the rank-one quadratic tests in (1.1) on every decomposable tangent–fiber direction. Positivity of the top Chern form is positivity of the scalar $F_n(\Phi)$ in (1.3), and (1.4) writes this scalar as $\mathbb{E}[\overline{\det G_n} \det \Phi(G_n)]$. Thus positivity of the map controls Φ on rank-one positive semidefinite matrices, whereas the sign of the top Chern form depends on the signed sum obtained by expanding both determinants. The expectation identifies the monomials in $\det \Phi(Y)$ that contribute to $F_n(\Phi)$. The construction makes their total contribution negative while preserving the rank-one inequalities in (1.1). Orthogonality and integration by parts express this contribution through weighted moments, which the Cayley identities evaluate.

1.3. Construction of the counterexample. The proof of Theorem 1.1 starts with a positive map for which $F_9 < 0$. The Gaussian representation identifies the terms responsible for the negative sign. A small trace perturbation makes the map strictly positive while preserving this sign. We then realize it as the curvature of a local metric and extend the metric to projective space, preserving its curvature near the origin and Griffiths positivity everywhere.

The positive three-dimensional reduction map

$$R : M_3 \rightarrow M_3, \quad R(A) := \operatorname{tr}(A)I_3 - A,$$

will be used below, and $F_3(R) = 0$. The six-dimensional Breuer–Hall map of Breuer [5] and Hall [16, Theorem 4] uses two copies of R and skew-symmetric off-diagonal blocks while preserving positivity. We append a three-dimensional diagonal block whose determinant weights the contribution of the two skew-symmetric blocks. Since $F_3(R) = 0$, the terms of F_9 independent of these blocks vanish, so their joint contribution alone determines the sign of F_9 ; this is what leads from the six-dimensional Breuer–Hall map to rank nine.

For the resulting family $\Phi_t : M_9 \rightarrow M_9$, $t \geq 0$, Gaussian integration by parts expresses that contribution through

$$\mathbb{E}[|\det G|^2 |\operatorname{tr} G|^2] - \mathbb{E}[|\det G|^2 \|G\|_{\mathbb{F}}^2],$$

where G is a 3×3 matrix with independent standard complex Gaussian entries and $\|G\|_{\mathbb{F}}^2 = \sum_{i,j} |g_{ij}|^2$. The trace involves the diagonal entries, whereas the Frobenius norm includes every entry. The Cayley identities evaluate these weighted moments and give

$$F_9(\Phi_t) = -1536t^3 \quad (t \geq 0).$$

Thus, for $t > 0$, the map satisfies $v^* \Phi_t(uu^*)v \geq 0$ for all u, v while the expectation in (1.4) is negative. By (1.4), its exact value is $-1536t^3$.

The same block determinant calculation extends the construction to every rank $r \geq 9$. Proposition 3.1 shows that adjoining scalar diagonal blocks and adding a small trace term produces a strictly positive map $\Lambda_r : M_r \rightarrow M_r$ with $F_r(\Lambda_r) < 0$.

The metric h_{Λ_r} on a neighborhood of the origin realizes the prescribed curvature there, as in Finski [11, Proposition 2.10]. After pairing with the metric, the curvature is the sum of positive rank-one evaluations of the map and a positive semidefinite correction, so the metric is Griffiths positive near the origin. The base-point formula (1.3) gives its top Chern form there as $F_r(\Lambda_r)dV_r$, which is negative.

At the origin, the local metric has the same value and first derivatives as the Fubini–Study direct sum metric on $\mathcal{O}_{\mathbb{P}^r}(1)^{\oplus r}$, so their difference vanishes to second order. A cutoff varying slowly on a logarithmic radial scale joins the two metrics in one step; the resulting curvature errors are dominated by the uniform Griffiths lower bound along the affine segment between

them. Proposition 3.2 gives this extension for the metric $I_q - 2\pi\Lambda(zz^*)$ associated with any strictly positive linear map $\Lambda : M_n \rightarrow M_q$.

Section 2 develops a positive-map formulation of the curvature at a point, represents the top Chern coefficient by a Gaussian moment, and constructs in rank nine a local Hermitian metric whose curvature is Griffiths semipositive and whose top Chern form is negative at the origin. Section 3 makes the underlying map strictly positive in every rank $r \geq 9$ and realizes its curvature by a globally Griffiths-positive metric on projective space, producing the compact counterexamples.

2. POSITIVE MAPS AND TOP CHERN FORMS

In this section, we establish the pointwise part of the construction: a Hermitian metric whose curvature is Griffiths semipositive at the origin and whose top Chern form is negative there. This construction will be used to obtain compact counterexamples.

For $n \geq 1$, write $M_n = M_n(\mathbb{C})$ for the space of $n \times n$ complex matrices, and let $\Phi : M_n \rightarrow M_n$ be a complex-linear map satisfying $\Phi(A^*) = \Phi(A)^*$ for every $A \in M_n$. On $\mathbb{C}^n$, use coordinates $z_j = x_j + iy_j$. For a Hermitian metric matrix h , we use the convention

$$\nabla = d + h^{-1}\partial h, \quad R^h = \bar{\partial}(h^{-1}\partial h), \quad \Omega^h = \frac{i}{2\pi}R^h. \quad (2.1)$$

Writing $h_i := \partial h / \partial z_i$ and similarly for the other derivatives gives

$$R_{i\bar{j}}^h = -\partial_{\bar{j}}(h^{-1}h_i) = h^{-1}h_{\bar{j}}h^{-1}h_i - h^{-1}h_{i\bar{j}}. \quad (2.2)$$

The Hermitian matrix

$$h_\Phi(z) = I_n - 2\pi\Phi(zz^*) \quad (2.3)$$

satisfies $h_\Phi(0) = I_n$ and is therefore positive definite near the origin. Its first derivatives satisfy $dh_\Phi(0) = 0$. With E_{ij} denoting the standard matrix units in M_n , its mixed second derivatives are $(h_\Phi)_{i\bar{j}} = -2\pi\Phi(E_{ij})$. Equation (2.2) therefore gives

$$\Omega^{h_\Phi}|_0 = -\frac{i}{2\pi} \sum_{i,j=1}^n (h_\Phi)_{i\bar{j}}(0) dz_i \wedge d\bar{z}_j|_0 = i \sum_{i,j=1}^n \Phi(E_{ij}) dz_i \wedge d\bar{z}_j|_0. \quad (2.4)$$

Use the positive volume form

$$dV_n = \bigwedge_{j=1}^n (i dz_j \wedge d\bar{z}_j) = 2^n dx_1 \wedge dy_1 \wedge \cdots \wedge dx_n \wedge dy_n. \quad (2.5)$$

Taking the determinant in (2.4) gives the complete expansion of the top Chern form at the origin:

$$\begin{aligned} c_n(\mathbb{C}^n \times \mathbb{C}^n, h_\Phi)|_0 &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \bigwedge_{a=1}^n \left(i \sum_{i_a, j_a=1}^n \Phi(E_{i_a j_a})_{a, \sigma(a)} dz_{i_a} \wedge d\bar{z}_{j_a} \right) \\ &= \sum_{\sigma, \rho, \eta \in S_n} \operatorname{sgn}(\sigma) \prod_{a=1}^n \Phi(E_{\rho(a), \eta(a)})_{a, \sigma(a)} \bigwedge_{a=1}^n (i dz_{\rho(a)} \wedge d\bar{z}_{\eta(a)}) \\ &= \sum_{\sigma, \rho, \eta \in S_n} \operatorname{sgn}(\sigma) \operatorname{sgn}(\rho) \operatorname{sgn}(\eta) \prod_{a=1}^n \Phi(E_{\rho(a), \eta(a)})_{a, \sigma(a)} dV_n|_0. \end{aligned}$$

Now let $Y = (y_{ij})$ be an $n \times n$ matrix of independent variables and write

$$\partial_Y := \left(\frac{\partial}{\partial y_{ij}} \right)_{i,j=1}^n.$$

Then

$$\begin{aligned} F_n(\Phi) &= \det(\partial_Y) \det \Phi(Y) \\ &= \sum_{\sigma, \tau \in S_n} \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau) \left(\prod_{i=1}^n \frac{\partial}{\partial y_{i, \tau(i)}} \right) \prod_{a=1}^n \left(\sum_{p,q=1}^n y_{pq} \Phi(E_{pq})_{a, \sigma(a)} \right) \\ &= \sum_{\sigma, \tau, \rho \in S_n} \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau) \prod_{a=1}^n \Phi(E_{\rho(a), \tau(\rho(a))})_{a, \sigma(a)} \\ &= \sum_{\sigma, \rho, \eta \in S_n} \operatorname{sgn}(\sigma) \operatorname{sgn}(\rho) \operatorname{sgn}(\eta) \prod_{a=1}^n \Phi(E_{\rho(a), \eta(a)})_{a, \sigma(a)}. \end{aligned}$$

Comparing the two expansions yields

$$c_n(\mathbb{C}^n \times \mathbb{C}^n, h_\Phi)|_0 = F_n(\Phi) dV_n|_0. \quad (2.6)$$

We call Φ positive if $\Phi(H) \geq 0$ for every $H \geq 0$. Here $A \geq 0$ means that A is a positive semidefinite Hermitian matrix. Equation (2.4) shows that Griffiths semipositivity of h_Φ at the origin is equivalent to $\Phi(uu^*) \geq 0$ for every $u \in \mathbb{C}^n$. Since every $H \geq 0$ can be written as $H = \sum_k u_k u_k^*$, this is equivalent to positivity of Φ . By (2.6), we therefore seek a positive map for which $F_n(\Phi) < 0$.

For $f, g \in \mathbb{C}[x_1, \dots, x_N]$, the apolar pairing is

$$\langle f, g \rangle := f(\partial_x)g(x)|_{x=0}. \quad (2.7)$$

With $x^\alpha = \prod_\nu x_\nu^{\alpha_\nu}$ and $\alpha! = \prod_\nu \alpha_\nu!$ for multi-indices $\alpha, \beta \in \mathbb{Z}_{\geq 0}^N$, this pairing satisfies

$$\langle x^\alpha, x^\beta \rangle = \begin{cases} \alpha!, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

The pairing is symmetric, and therefore

$$F_n(\Phi) = \langle \det Y, \det \Phi(Y) \rangle = \langle \det \Phi(Y), \det Y \rangle.$$

The next lemma realizes the apolar pairing as an expectation of polynomials in independent standard complex Gaussian variables and records the corresponding integration-by-parts identity.

Lemma 2.1. *Let $N \geq 1$ be an integer and $Z = (Z_1, \dots, Z_N)$ have independent standard complex Gaussian coordinates, each with density $\pi^{-1}e^{-|z|^2}$. For polynomials $f, g, q \in \mathbb{C}[x_1, \dots, x_N]$, the following identities hold.*

(i) *The Gaussian realization of the apolar pairing is*

$$\langle f, g \rangle = \mathbb{E}[f(Z)g(\bar{Z})]. \quad (2.8)$$

(ii) *For every $1 \leq \nu \leq N$, Gaussian integration by parts gives*

$$\mathbb{E}[\bar{Z}_\nu f(Z)g(\bar{Z})] = \mathbb{E}[(\partial_{x_\nu} f)(Z)g(\bar{Z})]. \quad (2.9)$$

Consequently, multiplication transfers to differentiation as

$$\langle f, qg \rangle = \langle q(\partial_x) f, g \rangle. \quad (2.10)$$

Proof. (i) Angular integration and the substitution $t = |z|^2$ give, for nonnegative integers a, b ,

$$\mathbb{E}[Z_\nu^a \overline{Z}_\nu^b] = \begin{cases} 0, & a \neq b, \\ \int_0^\infty t^a e^{-t} dt = a!, & a = b. \end{cases}$$

Independence gives the corresponding identity for monomials in $Z_1, \dots, Z_N$, and bilinearity proves (2.8) for arbitrary complex coefficients.

(ii) Write $z_\nu = u_\nu + iv_\nu$. Since

$$\overline{z}_\nu e^{-\|z\|^2} = -\frac{1}{2}(\partial_{u_\nu} - i\partial_{v_\nu})e^{-\|z\|^2}, \quad (\partial_{u_\nu} - i\partial_{v_\nu})(f(z)g(\overline{z})) = 2(\partial_{x_\nu} f)(z)g(\overline{z}),$$

ordinary integration by parts gives

$$\begin{aligned} \mathbb{E}[\overline{Z}_\nu f(Z)g(\overline{Z})] &= -\frac{1}{2\pi^N} \int_{\mathbb{C}^N} f(z)g(\overline{z})(\partial_{u_\nu} - i\partial_{v_\nu})e^{-\|z\|^2} d\lambda(z) \\ &= \frac{1}{2\pi^N} \int_{\mathbb{C}^N} (\partial_{u_\nu} - i\partial_{v_\nu})(f(z)g(\overline{z}))e^{-\|z\|^2} d\lambda(z) \\ &= \mathbb{E}[(\partial_{x_\nu} f)(Z)g(\overline{Z})]. \end{aligned}$$

Here λ is Lebesgue measure on $\mathbb{C}^N$. Applying (2.9) to the factors of each monomial in q and using linearity gives

$$\mathbb{E}[f(Z)q(\overline{Z})g(\overline{Z})] = \mathbb{E}[(q(\partial_x)f)(Z)g(\overline{Z})].$$

Together with (2.8), this proves (2.10). $\square$

If $h \in \mathbb{C}[y_{ij} : 1 \leq i, j \leq n]$ is homogeneous of degree n , then $\det(\partial_Y)h$ is constant. The symmetry of the apolar pairing, the real coefficients of the determinant, and Lemma 2.1(i) give

$$\det(\partial_Y)h = \langle h, \det Y \rangle = \mathbb{E}[h(G_n)\overline{\det G_n}].$$

Here G_n is an $n \times n$ random matrix with independent standard complex Gaussian entries. Taking $h = \det \Phi(Y)$ gives

$$F_n(\Phi) = \mathbb{E}[\overline{\det G_n} \det \Phi(G_n)],$$

as in (1.4). In this expectation, a monomial has zero expectation whenever some Gaussian variable and its conjugate occur with different powers. This orthogonality identifies the terms that can contribute to F_n ; we seek a positive map for which their sum is negative.

The rank-nine map is built from the six-dimensional Breuer–Hall map of Breuer [5] and Hall [16, Theorem 4], which takes two copies of the three-dimensional reduction map together with skew-symmetric off-diagonal blocks and is positive. We enlarge it by a three-dimensional diagonal block whose determinant occurs as a factor in the determinant of the whole output. In the resulting rank-nine map, $F_3(R) = 0$ eliminates the terms independent of the skew-symmetric blocks, so the negative Gaussian expectation comes from those blocks.

Write $\mathcal{Y} \in M_9(\mathbb{C})$ as the following 3×3 matrix of 3×3 blocks:

$$\mathcal{Y} = \begin{pmatrix} X & W & U_{13} \\ V & Y & U_{23} \\ U_{31} & U_{32} & Z \end{pmatrix}, \quad S := \begin{pmatrix} X & W \\ V & Y \end{pmatrix}.$$

For $J := \begin{pmatrix} 0 & I_3 \\ -I_3 & 0 \end{pmatrix}$, consider

$$B_6(S) := \operatorname{tr}(S)I_6 - S - JS^\top J^*. \quad (2.11)$$

The map B_6 is the six-dimensional Breuer–Hall map. We now consider the following family of maps on $M_9(\mathbb{C})$:

$$\Phi_t(\mathcal{Y}) := \begin{pmatrix} \begin{pmatrix} Z & 0 \\ 0 & Z^\top \end{pmatrix} + tB_6(S) & 0 \\ 0 & X + Y^\top \end{pmatrix}, \quad t \geq 0. \quad (2.12)$$

Transposition in (2.11) and (2.12) acts on matrix indices and does not conjugate scalars, so both maps are complex-linear.

Proposition 2.2. *For every real $t \geq 0$, the map Φ_t in (2.12) is positive.*

Proof. The matrix J satisfies $J^\top = -J$ and $J^*J = I_6$. Hence Hall’s positivity theorem for the extended reduction map [16, Theorem 4] applies to (2.11) and gives $B_6(S) \geq 0$ whenever $S \geq 0$.

If $\mathcal{Y} \geq 0$, then $S, X, Y, Z \geq 0$. Together with $B_6(S) \geq 0$ and $t \geq 0$, these inequalities prove $\Phi_t(\mathcal{Y}) \geq 0$. $\square$

Proposition 2.2 and (2.4) show that $\Omega^{h_{\Phi_t}}|_0$ is Griffiths semipositive. By (2.6),

$$c_9(\mathbb{C}^9 \times \mathbb{C}^9, h_{\Phi_t})|_0 = F_9(\Phi_t)dV_9|_0.$$

The desired separation of the two positivity conditions therefore follows from the next exact evaluation.

Theorem 2.3. *For every real $t \geq 0$, the positive map Φ_t of (2.12) satisfies*

$$F_9(\Phi_t) = -1536t^3. \quad (2.13)$$

Proof. Set

$$A := X + Y^\top, \quad K := W^\top - W, \quad L := V^\top - V.$$

A direct computation gives

$$JS^\top J^* = \begin{pmatrix} Y^\top & -W^\top \\ -V^\top & X^\top \end{pmatrix}, \quad \text{tr}(S) = \text{tr}(A),$$

and therefore

$$B_6(S) = \begin{pmatrix} R(A) & K \\ L & R(A)^\top \end{pmatrix}, \quad \Phi_t(\mathcal{Y}) = \begin{pmatrix} \begin{pmatrix} Z & 0 \\ 0 & Z^\top \end{pmatrix} + tB_6(S) & 0 \\ 0 & A \end{pmatrix}, \quad (2.14)$$

where $R(A) = \text{tr}(A)I_3 - A$ is the three-dimensional reduction map on $M_3(\mathbb{C})$. Consequently,

$$\det \Phi_t(\mathcal{Y}) = \det A \det \begin{pmatrix} Z + tR(A) & tK \\ tL & Z^\top + tR(A)^\top \end{pmatrix}.$$

For a column vector $a = (a_1, a_2, a_3)^\top$, the notation $[a]$ denotes the matrix

$$[a] := \begin{pmatrix} 0 & a_3 & -a_2 \\ -a_3 & 0 & a_1 \\ a_2 & -a_1 & 0 \end{pmatrix}. \quad (2.15)$$

Write $W = (w_{ij})_{i,j=1}^3$ and $V = (v_{ij})_{i,j=1}^3$. Under the convention (2.15), the skew-symmetric matrices $K = W^\top - W$ and $L = V^\top - V$ are represented by the column vectors

$$k := \begin{pmatrix} w_{32} - w_{23} \\ w_{13} - w_{31} \\ w_{21} - w_{12} \end{pmatrix}, \quad l := \begin{pmatrix} v_{32} - v_{23} \\ v_{13} - v_{31} \\ v_{21} - v_{12} \end{pmatrix}. \quad (2.16)$$

Indeed, $[k] = K = W^\top - W$ and $[l] = L = V^\top - V$. Set $M := Z + tR(A)$. If M is invertible, then (2.15) gives

$$M^{-\top}[l]M^{-1} = \frac{1}{\det M}[Ml], \quad [Ml][k] = k(Ml)^\top - (k^\top Ml)I_3.$$

The Schur complement and the matrix determinant lemma give

$$\begin{aligned} \det \Phi_t(\mathcal{Y}) &= (\det A)(\det M)^2 \det \left(I_3 - \frac{t^2}{\det M}[Ml][k] \right) \\ &= (\det A)(\det M + t^2 k^\top Ml)^2 \\ &= (\det A) \left[\det(Z + tR(A)) + t^2 k^\top (Z + tR(A))l \right]^2. \end{aligned} \quad (2.17)$$

Both sides are polynomials in the entries of M , so the identity also holds when M is singular.

The polynomial in (2.17) depends only on $\mathcal{Y}_{ij}$ with either $1 \leq i, j \leq 6$ or $7 \leq i, j \leq 9$. Consequently, a term $\prod_{i=1}^9 \partial/\partial \mathcal{Y}_{i,\sigma(i)}$ in $\det(\partial_{\mathcal{Y}})$ vanishes unless σ preserves both $\{1, \dots, 6\}$ and $\{7, 8, 9\}$. Thus only $\sigma \in S_6 \times S_3 \subset S_9$ contribute, and on this polynomial

$$\det(\partial_{\mathcal{Y}}) = \det(\partial_S) \det(\partial_Z). \quad (2.18)$$

The polynomial $\det \Phi_t(\mathcal{Y})$ is homogeneous of total degree nine in the entries of S and Z . Since $\det(\partial_Z)$ is a differential operator of order three, $\det(\partial_Z) \det \Phi_t(\mathcal{Y})$ is homogeneous of total degree six in the entries of S and Z . Every term of positive degree in Z has degree at most five in the entries of S and is therefore annihilated by the differential operator $\det(\partial_S)$, which has order six. Therefore,

$$F_9(\Phi_t) = \det(\partial_S) \det(\partial_Z) \det \Phi_t(\mathcal{Y}) = \det(\partial_S) (\det(\partial_Z) \det \Phi_t(\mathcal{Y})|_{Z=0}). \quad (2.19)$$

The ordinary and all-minors Cayley identities for 3×3 matrices [6, Theorem 2.1] give

$$\det(\partial_M)((\det M)^2) = 24 \det M, \quad \det(\partial_M)((\det M)M_{ij}) = 8M_{ij}. \quad (2.20)$$

The second formula follows by applying the product rule to the cofactor case of the all-minors identity. Expanding the square now yields, for $a, b \in \mathbb{C}^3$,

$$\det(\partial_M)(\det M + a^\top Mb)^2 = 24 \det M + 16a^\top Mb.$$

Taking $M = Z + tR(A)$, $a = tk$, and $b = tl$ gives

$$\begin{aligned} \det(\partial_Z) \det \Phi_t(\mathcal{Y})|_{Z=0} &= (\det A) \left[24 \det(Z + tR(A)) + 16t^2 k^\top (Z + tR(A))l \right] \Big|_{Z=0} \\ &= t^3 (\det A) [24 \det R(A) + 16k^\top R(A)l]. \end{aligned} \quad (2.21)$$

To apply the remaining factor $\det(\partial_S)$ in (2.18) to (2.21), write

$$\mathcal{D} := \det(\partial_A), \quad \mathcal{L} := \sum_{i,j=1}^3 \frac{\partial}{\partial A_{ij}} \frac{\partial}{\partial k_i} \frac{\partial}{\partial l_j}. \quad (2.22)$$

On polynomials in the scalar variables A_{ij}, k_r, l_r ($1 \leq i, j, r \leq 3$), regarded as polynomials in the entries of X, Y, W, V through $A = X + Y^\top$ and (2.16), one has

$$\partial_X = \partial_A, \quad \partial_Y = \partial_A^\top, \quad \partial_W = -[\partial_k], \quad \partial_V = -[\partial_l]. \quad (2.23)$$

All these constant-coefficient differential operators commute. The block determinant calculation in (2.17) therefore gives

$$\begin{aligned}
\det(\partial_S) &= \det \begin{pmatrix} \partial_A & -[\partial_k] \\ -[\partial_l] & \partial_A^\top \end{pmatrix} \\
&= \left(\det(\partial_A) + (-\partial_k)^\top \partial_A (-\partial_l) \right)^2 \\
&= \left(\det(\partial_A) + \sum_{i,j=1}^3 \frac{\partial}{\partial A_{ij}} \frac{\partial}{\partial k_i} \frac{\partial}{\partial l_j} \right)^2 \\
&= (\mathcal{D} + \mathcal{L})^2 = \mathcal{D}^2 + 2\mathcal{D}\mathcal{L} + \mathcal{L}^2.
\end{aligned} \tag{2.24}$$

Since $(\det A) \det R(A)$ is independent of k, l ,

$$\mathcal{L}((\det A) \det R(A)) = 0, \quad (\mathcal{D} + \mathcal{L})^2((\det A) \det R(A)) = \mathcal{D}^2((\det A) \det R(A)).$$

For the second summand, writing $k^\top R(A)l = \sum_{p,q=1}^3 k_p R(A)_{pq} l_q$ gives

$$\mathcal{L}((\det A) k^\top R(A)l) = \sum_{i,j,p,q=1}^3 \partial_{A_{ij}} \partial_{k_i} \partial_{l_j} ((\det A) k_p R(A)_{pq} l_q) = \sum_{i,j=1}^3 \partial_{A_{ij}} ((\det A) R(A)_{ij}).$$

Moreover, $(\det A) k^\top R(A)l$ has degree four in the entries of A , whereas $\mathcal{D}^2$ has order six in those entries. Consequently,

$$(\mathcal{D} + \mathcal{L})^2((\det A) k^\top R(A)l) = 2\mathcal{D} \sum_{i,j=1}^3 \partial_{A_{ij}} ((\det A) R(A)_{ij}).$$

By (2.19), (2.21), and (2.24),

$$\begin{aligned}
F_9(\Phi_t) &= t^3 (\mathcal{D} + \mathcal{L})^2 \left((\det A) [24 \det R(A) + 16 k^\top R(A)l] \right) \\
&= t^3 \left[24 \mathcal{D}^2((\det A) \det R(A)) + 32 \mathcal{D} \sum_{p,q=1}^3 \partial_{A_{pq}} ((\det A) R(A)_{pq}) \right].
\end{aligned} \tag{2.25}$$

We next compute in turn the two terms in (2.25). Writing $\tau := \operatorname{tr} A$, the characteristic polynomial of A gives

$$\det R(A) = \tau e_2(A) - \det A, \quad e_2(A) = \sum_{1 \leq i < j \leq 3} (A_{ii} A_{jj} - A_{ij} A_{ji}).$$

A direct evaluation gives $\mathcal{D}(\tau e_2(A)) = \mathcal{D}(\det A) = 6$, and hence $\mathcal{D} \det R(A) = 0$. Equations (2.10) and (2.20) now give

$$\begin{aligned}
\mathcal{D}^2((\det A) \det R(A)) &= \langle (\det A)^2, (\det A) \det R(A) \rangle \\
&= \langle \mathcal{D}((\det A)^2), \det R(A) \rangle = 24 \mathcal{D} \det R(A) = 0.
\end{aligned} \tag{2.26}$$

For the second term in (2.25), let $G = (g_{pq})$ be a 3×3 random matrix with independent standard complex Gaussian entries. By Lemma 2.1, one has

$$\begin{aligned}
\mathcal{D} \sum_{p,q} \partial_{A_{pq}} ((\det A) R(A)_{pq}) &= \sum_{p,q} \mathbb{E} [\overline{\det G} \partial_{A_{pq}} ((\det A) R(A)_{pq})(G)] \\
&= \mathbb{E} [|\det G|^2 \sum_{p,q} \overline{g_{pq}} R(G)_{pq}] \\
&= \mathbb{E} [|\det G|^2 (|\operatorname{tr} G|^2 - \|G\|_{\mathbb{F}}^2)].
\end{aligned} \tag{2.27}$$

Here $\|G\|_F^2 = \sum_{p,q} |g_{pq}|^2$. Lemma 2.1(i) and (2.20) yield, for $1 \leq p, q, r, s \leq 3$,

$$\begin{aligned} \mathbb{E}[|\det G|^2 g_{pq} \overline{g_{rs}}] &= \langle (\det A) A_{rs}, (\det A) A_{pq} \rangle \\ &= \partial_{A_{rs}} \mathcal{D}((\det A) A_{pq})|_{A=0} = 8\delta_{pr}\delta_{qs}. \end{aligned}$$

Thus the diagonal cross terms in the weighted trace moment vanish, while every squared entry contributes to the weighted Frobenius moment:

$$\mathbb{E}[|\det G|^2 |\operatorname{tr} G|^2] = 24, \quad \mathbb{E}[|\det G|^2 \|G\|_F^2] = 72.$$

Their difference determines the sign of the contribution from the two skew-symmetric blocks in (2.27):

$$\mathbb{E}[|\det G|^2 (|\operatorname{tr} G|^2 - \|G\|_F^2)] = -48. \quad (2.28)$$

Equations (2.25)–(2.28) yield

$$F_9(\Phi_t) = -1536t^3,$$

which proves (2.13). $\square$

3. COMPACT COUNTEREXAMPLES FROM POSITIVE MAPS

We extend the rank-nine map to every $r \geq 9$, make it strictly positive by a trace perturbation, and realize the resulting map on projective space.

For $r \geq 9$, $\eta \geq 0$, and $Y = (y_{ij})_{i,j=1}^r \in M_r$, let X be the leading 9×9 principal submatrix of Y and set

$$\Lambda_{r,\eta}(Y) := \operatorname{diag}(\Phi_1(X), y_{10,10}, \dots, y_{rr}) + \eta \operatorname{tr}(Y) I_r, \quad (3.1)$$

where the scalar diagonal entries are omitted when $r = 9$.

Proposition 3.1. *For every integer $r \geq 9$, there exists $\eta_0 = \eta_0(r) > 0$ such that $\Lambda_{r,\eta}$ is strictly positive and $F_r(\Lambda_{r,\eta}) < 0$ whenever $0 < \eta < \eta_0$.*

Proof. If $Y \geq 0$, then $X \geq 0$ and $y_{jj} \geq 0$ for $j > 9$. Proposition 2.2 therefore shows that $\Lambda_{r,0}$ is positive. Moreover,

$$\det \Lambda_{r,0}(Y) = \det \Phi_1(X) \prod_{j=10}^r y_{jj}.$$

By (1.2),

$$F_r(\Lambda_{r,0}) = \det(\partial_X) \det \Phi_1(X) = F_9(\Phi_1) = -1536.$$

For every nonzero $Y \geq 0$ and every $\eta > 0$,

$$\Lambda_{r,\eta}(Y) = \Lambda_{r,0}(Y) + \eta \operatorname{tr}(Y) I_r \geq \eta \operatorname{tr}(Y) I_r > 0.$$

Thus $\Lambda_{r,\eta}$ is strictly positive. Its coefficients are real, so $F_r(\Lambda_{r,\eta})$ is a real polynomial in η . Since its value at $\eta = 0$ is -1536 , it remains negative for all sufficiently small $\eta > 0$. $\square$

For a strictly positive complex-linear map $\Lambda : M_n \rightarrow M_q$, (2.3) takes the form

$$h_\Lambda(z) := I_q - 2\pi\Lambda(zz^*).$$

The following proposition realizes its curvature at a point by a global Griffiths-positive metric. The realization can be confined to an arbitrary neighborhood of that point.

Proposition 3.2. *Let $n, q \geq 1$ and $\Lambda : M_n \rightarrow M_q$ be a strictly positive complex-linear map. The bundle $\mathcal{O}_{\mathbb{P}^n}(1)^{\oplus q}$ admits a smooth Griffiths-positive Hermitian metric h such that, in the standard affine coordinates and local frame at $p := [1 : 0 : \cdots : 0]$,*

$$\Omega^h|_p = i \sum_{i,j=1}^n \Lambda(E_{ij}) dz_i \wedge d\bar{z}_j|_p.$$

Given any neighborhood U of p , the metric can be chosen to agree with the Fubini–Study direct sum metric outside U and with h_Λ near p .

Proof. Fix a neighborhood U of p . Strict positivity and compactness of the product of unit spheres give a constant $a > 0$ such that

$$\Lambda(uu^*) \geq a\|u\|^2 I_q \quad (u \in \mathbb{C}^n).$$

On the standard affine chart, write $s := \|z\|^2$ and let

$$h_0(z) := \frac{I_q}{1+s}, \quad h_1(z) := h_\Lambda(z), \quad h_t := (1-t)h_0 + th_1 \quad (0 \leq t \leq 1).$$

For a constant tangent vector u , subscripts $u, \bar{u}$ denote directional derivatives. At the origin,

$$h_t(0) = I_q, \quad -(h_t)_{u\bar{u}}(0) = (1-t)\|u\|^2 I_q + 2\pi t \Lambda(uu^*) \geq \min\{1, 2\pi a\} \|u\|^2 I_q.$$

Continuity and compactness in t and in the unit tangent and fiber vectors allow us to choose a coordinate ball B_ρ with $\bar{B}_\rho \subset U$ and a constant $c > 0$ such that, throughout $\bar{B}_\rho$,

$$h_t \geq cI_q, \quad -(h_t)_{u\bar{u}} \geq c\|u\|^2 I_q \quad (0 \leq t \leq 1). \quad (3.2)$$

Set $H := h_1 - h_0$ and $\sigma := z^*u$. Direct differentiation gives

$$H(z) = \frac{s}{1+s} I_q - 2\pi \Lambda(zz^*), \quad H_u(z) = \frac{\sigma}{(1+s)^2} I_q - 2\pi \Lambda(uz^*).$$

Every linear map between finite-dimensional normed spaces is bounded. Hence, for a constant $C > 0$ independent of z and u ,

$$\|\Lambda(zz^*)\|_{\text{op}} \leq Cs, \quad \|\Lambda(uz^*)\|_{\text{op}} \leq C\sqrt{s}\|u\|.$$

Together with $|\sigma| \leq \sqrt{s}\|u\|$, these estimates give, for some constant $C' > 0$,

$$\begin{aligned} \|H(z)\|_{\text{op}} &\leq \frac{s}{1+s} + 2\pi \|\Lambda(zz^*)\|_{\text{op}} \leq C's, \\ \|H_u(z)\|_{\text{op}} &\leq \frac{|\sigma|}{(1+s)^2} + 2\pi \|\Lambda(uz^*)\|_{\text{op}} \leq C'\sqrt{s}\|u\| \quad (z \in \bar{B}_\rho). \end{aligned}$$

Choose a smooth function $\theta : \mathbb{R} \rightarrow [0, 1]$ which equals one on $(-\infty, 0]$ and zero on $[1, \infty)$. Given $L \geq 1$, write

$$\chi_L(z) := \begin{cases} \theta\left(1 + \frac{1}{L} \log \frac{s}{\rho^2}\right), & z \neq 0, \\ 1, & z = 0. \end{cases}$$

It is smooth, equals one when $\|z\| \leq \rho e^{-L/2}$, and vanishes when $\|z\| \geq \rho$. At $z \neq 0$, one has

$$(\chi_L)_u = \frac{\theta'(\tau)\sigma}{Ls}, \quad (\chi_L)_{u\bar{u}} = \frac{\theta''(\tau)|\sigma|^2}{L^2s^2} + \frac{\theta'(\tau)}{L} \left(\frac{\|u\|^2}{s} - \frac{|\sigma|^2}{s^2} \right),$$

where $\tau := 1 + L^{-1} \log(s/\rho^2)$. Since $|\sigma|^2 \leq s\|u\|^2$, there is a constant $C'' > 0$, depending only on θ and independent of L, z, u , such that

$$|(\chi_L)_u| \leq \frac{C''}{L\sqrt{s}}\|u\|, \quad |(\chi_L)_{u\bar{u}}| \leq \frac{C''}{Ls}\|u\|^2.$$

On B_ρ , take

$$h := h_0 + \chi_L H = (1 - \chi_L)h_0 + \chi_L h_1.$$

The first inequality in (3.2) shows that h is positive definite. Differentiating h gives

$$-h_{u\bar{u}} = -(1 - \chi_L)(h_0)_{u\bar{u}} - \chi_L(h_1)_{u\bar{u}} - (\chi_L)_{u\bar{u}}H - (\chi_L)_u H_{\bar{u}} - (\chi_L)_{\bar{u}} H_u.$$

Since $H_{\bar{u}} = H_u^*$, the sum of the last three terms is a Hermitian matrix whose operator norm is at most $3C'C''L^{-1}\|u\|^2$. Choosing L so that $3C'C''/L < c/2$, the second inequality in (3.2) gives

$$-h_{u\bar{u}} \geq \frac{c}{2}\|u\|^2 I_q.$$

Since $\chi_L \equiv 1$ on a neighborhood of 0, one has $h = h_1$ there. Hence the inequality also holds at $z = 0$. Equation (2.2) then gives

$$hR^h(u, \bar{u}) = -h_{u\bar{u}} + h_u^* h^{-1} h_u \geq \frac{c}{2}\|u\|^2 I_q,$$

so h is Griffiths positive on B_ρ .

Because $\chi_L = 0$ for $\|z\| \geq \rho$, h extends smoothly to a global Griffiths-positive metric that agrees with the Fubini–Study direct sum metric outside U . Near p , h coincides with $h_1 = h_\Lambda$. In particular, $h(p) = I_q$, $dh(p) = 0$, and $h_{i\bar{j}}(p) = -2\pi\Lambda(E_{ij})$, so (2.2) gives

$$\Omega^h|_p = -\frac{i}{2\pi} \sum_{i,j=1}^n h_{i\bar{j}}(p) dz_i \wedge d\bar{z}_j|_p = i \sum_{i,j=1}^n \Lambda(E_{ij}) dz_i \wedge d\bar{z}_j|_p.$$

□

Proof of Theorem 1.1. Fix $r \geq 9$, choose $0 < \eta < \eta_0(r)$ as in Proposition 3.1, and set $\Lambda_r := \Lambda_{r,\eta}$. Then Λ_r is strictly positive and $F_r(\Lambda_r) < 0$. Proposition 3.2 gives a smooth Griffiths-positive metric h_r on $E_r = \mathcal{O}_{\mathbb{P}^r}(1)^{\oplus r}$ whose normalized curvature at $p = [1 : 0 : \cdots : 0]$ is

$$\Omega^{h_r}|_p = i \sum_{i,j=1}^r \Lambda_r(E_{ij}) dz_i \wedge d\bar{z}_j|_p.$$

Taking the determinant of this curvature matrix, as in (2.6), gives

$$c_r(E_r, h_r)|_p = F_r(\Lambda_r) dV_r|_p < 0.$$

By continuity, the top Chern form is negative on a neighborhood of p .

□

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AI USE DISCLOSURE

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